

Linke Fan

21-year-old, Male, +86-13934242828, linke@tongji.edu.cn

fanlinke.github.io | github.com/fanlinke

EDUCATION

09/2023-06/2027	Tongji University (Shanghai, China)	Bachelor of Engineering
◆ School of Computer Science and Technology ◆ Major: Computer Science and Technology ◆ Grade Average: 85.04/100 ◆ Selected Coursework: <i>Information Processing (98)</i>, <i>Principles of Database Systems (95)</i>, <i>Computer Architecture (93)</i>, <i>Machine Learning (89)</i>, <i>Algorithm Analysis and Design (87)</i>. ◆ Attended the Plenary Session on AI Development and Safety at the 2025 World Artificial Intelligence Conference, July 2025 as an Outstanding Student Representative.		

PUBLICATION

Fan, L., Li, X., Huang, H., & Shu, K. (2026). FSB-Net: Frequency-Spatial Boundary Network for Brain Stroke Lesion Segmentation in Non-Contrast CT. *Proceedings of the 2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026, CCF-B)*. In press.

RESEARCH PROJECTS

11/2025-06/2026 **FSB-Net: Frequency-Spatial Boundary Network for Brain Stroke Lesion Segmentation in Non-Contrast CT**
Supervised by Dr. Kai Shu

- ◆ Developed FSB-Net for stroke lesion segmentation in non-contrast CT, using high-frequency information to improve the modelling of unclear lesion boundaries.
- ◆ Built a wavelet-based boundary detection head using discrete wavelet transforms on multi-scale features, and introduced a frequency-spatial cross-attention module to fuse boundary representations with spatial decoder features.
- ◆ Designed a spectral boundary loss in the Fourier domain to reduce high-frequency discrepancies and improve lesion boundary reconstruction, and evaluated its effectiveness through comparative experiments.

09/2025-07/2026 **CausIDL: Resource Competition-Aware Causal Graph Discovery for Deep Learning Microservices|**
Project lead

Supervised by: *Professor Zhijun Ding, Dr Binbin Feng*

- ◆ Investigated the limitations of applying conventional causal discovery methods to deep learning microservices, particularly when multiple instances compete for shared GPU, CPU, and memory resources.
- ◆ Built an instance-level causal discovery framework using GPU, CPU, memory, request, error, and latency metrics under dynamic service scaling.
- ◆ Used node-level aggregate metrics as shared parent nodes to model resource competition among service instances.
- ◆ Designed a domain knowledge-guided pruning mechanism that encodes GPU hardware characteristics and inference-service logic as causal priors, reducing unnecessary causal search through a forbidden-edge list.

12/2023-12/2024 **Serverless End-to-End Automated AI Workflow System| Research team member**

Supervised by: *Professor Zhijun Ding*

- ◆ Contributed to the design and development of a serverless system for automating end-to-end AI workflows.
- ◆ Developed core modules including an online AI code editor and a function-level parser that identifies dependencies and automatically generates computational graphs for workflow execution.
- ◆ Integrated and tested workflow execution through Argo, AI training and inference functions on Knative, and Kubernetes–Prometheus monitoring for Pod status and resource metrics.
- ◆ Tested the end-to-end workflow and contributed to a working prototype covering code development, dependency parsing, workflow execution, and system monitoring.

INTERNSHIP EXPERIENCES

08/2025-09/2025 JD Technology Group (Beijing, China)

Position: *Data Platform Product R&D Intern, IaaS Data Platform of Cloud Division*

- ◆ Supported an end-to-end cloud data lake migration project, addressing the challenges of transferring large-scale partitioned data reliably across cloud environments.
- ◆ Developed and optimised automated migration scripts for batch data transfer, reducing manual operations across multi-partition tables.
- ◆ Worked with the team on Spark and Hive performance analysis and query optimisation to address big-data query latency.
- ◆ Built databases and tables on JD Cloud, configured cloud server environments, and validated migrated data through queries to ensure integrity and consistency.

07/2025-08/2025 QAX Technology Group (Shanghai, China)

Position: *Pre-Sales Technical Intern, Bank Sales Division*

- ◆ Analysed cybersecurity requirements for banking clients through needs research and client interviews, translating business needs into structured technical requirements and documentation.
- ◆ Researched cybersecurity solutions and product capabilities, mapping technical features to client scenarios and contributing to solution design, technical proposals, and product documentation.
- ◆ Participated in technical discussions and solution demonstrations, documented technical feedback, and consolidated technical materials for project bids.

07/2024-08/2024 CCB FinTech Co., Ltd. (Shanghai, China)

Position: *Intern, Production & Technology Committee Office*

- ◆ Analysed production incident and operational risk data, prepared root-cause analysis reports, and tracked corrective actions to support system stability and risk control.
- ◆ Participated in core-system solution reviews and testing, consolidating technical findings and supporting project-wide risk assessment and implementation oversight.

ACTIVITIES & LEADERSHIP

09/2025-06/2026 Director, Youth League Organisation Department, School of Computer Science & Technology

- ◆ Managed Youth League membership development and administration, organised thematic activities, and coordinated leadership training for newly appointed class secretaries.

12/2025 Director & Event Lead, New Year Gala of the School of Computer Science and Technology

- ◆ Curated 11 cross-disciplinary performances with AI-interactive and code-based visual elements, coordinating 100+ performers and support staff to ensure smooth execution.

STANDARDISED TESTS

IELTS Test Date: 03/28/2026 Score: 7.5

SKILLS & CERTIFICATE

Certificate: Certificate of Completion in the 2026 Tencent AI Arena Artificial Intelligence Platform Professional Practice

- Technicals:**
- ◆ **Programming:** Python, C++, SQL
 - ◆ **AI & Data:** PyTorch, NumPy, Pandas
 - ◆ **Cloud-Native:** Kubernetes, Knative, Argo, Prometheus, Serverless Computing
 - ◆ **Big Data:** Spark, Hive, HDFS, Hudi